Module_5_(Alzheimer's Disease)

Team Members:

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Project Title

Heat Injury and Alzheimer's Disease

Project Goal

We are seeking to computationally and statistically analyze real patient data to gain a deeper understanding about Alzheimer's disease and how it is influenced by previous heat injuries.

Disease Background:

Info & references about Alzheimer's:

- Prevalence & Incidence:
  - 6.7 million people in the United States - 1.5 million people with Alzheimer's disease in the U.S. all groups (1.25% of the total U.S. population) incidence rate - 500,000 new cases annually 0.75% per year
  - Survival: Average 8 years (Mortality) - 1 patient per 65, 100, per 200,000,000, 100,000 in U.S., 1.5 million (Institute of Physician - 3 million in the 1990 Medicare and medicaid cost - $226 billion per year - 10% of total cost Costs: Hospital bed, nursing home/rehabilitation, long-term care, patients leaving institutions, home modifications & equipment (CHAPTO)
- Risk factors (genetic, lifestyle)

from Chatgpt

- Genetic Age - The strongest known risk factor. Risk doubles every 5 years after age 65 Family History - Presence in siblings with Alzheimer Increases risk Genes: Risk Genes - APOE ε4 is the most significant known risk gene, carrying one copy increases risk, and two copies raise it further. Deterministic Genes - Mutations in APP, PSEN1, and PSEN2 almost always lead to Alzheimer's. Lifestyle and Health Related Risk Factors Cardiovascular Health - What's good for the heart is good for the brain. Hypertension, high cholesterol, diabetes, obesity, smoking, and sedentary lifestyle increase risk. Diet - Mediterranean or MIND diet (rich in antioxidants, omega-3 fatty acids, low in saturated fat) slows down cognitive decline. Heavy alcohol use is hazardous. Sleep Problems - Chronic sleep deprivation and sleep apnea are linked to buildup of amyloid plaques. Cognitive & Social Engagement - Limited mental stimulation, social isolation, and low education increase risk. Brain Health Markers - Mild Cognitive Impairment (MCI) - Memory problems beyond normal aging, often a precursor to dementia. Biomarkers
  - Molecular or screen-level tests Depression and Stress - Provides clue to others or brain mechanisms and inflammation. Environmental Factors - Exposure to air pollution, toxins, or heavy metals
- Societal Alzheimer's Risk Factors - Low Income Lower education - education challenges the brain, building a cognitive reserve that protects the brain from damage Black / Hispanic populations - Due to healthcare disparities, higher risk factors, and systemic inequities. Caregiver Burden - Not a direct risk factor, but highlights the disease's impact on families and healthcare systems. Access to healthcare - Limited access to screening, diagnosis, and treatment exacerbates outcomes. Lack of preventive care and support - Lack of preventive care and support contributes to cognitive decline and increased risk. Cognitive reserve - Education, occupation, and lifelong learning can delay the onset of dementia, resulting in a longer life lived fully. Cognitive stimulation - Challenging the brain through learning and mental activity strengthens neural pathways and boosts cognitive resilience. Heavy alcohol use is hazardous. Sleep Problems - Chronic sleep deprivation and sleep apnea are linked to buildup of amyloid plaques. Cognitive & Social Engagement - Limited mental stimulation, social isolation, and low education increase risk. Brain Health Markers - Mild Cognitive Impairment (MCI) - Memory problems beyond normal aging, often a precursor to dementia. Biomarkers
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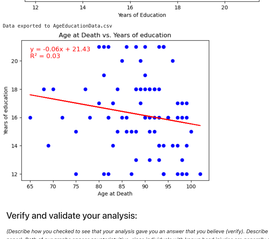
Sources:

```
    def get_thal(self):
        return self.thal_score

    def get_death_age(self):
        return self.death_age

    def get_age_diag(self):
        return self.age_diag

    @classmethod
    def combine_data(cls, filename: str):
        with open(filename, encoding="utf8") as f:
            reader = csv.DictReader(f)
            rows_of_patients = list(reader)
            #for line in csv create object
            for row in range(len(rows_of_patients)):
                if Patient.all_patients[row].DonorID == rows_of_patients[row]["Donor ID"]:
                    if rows_of_patients[row]["Sex"] != "":
                        Patient.all_patients[row].sex = rows_of_patients[row]["Sex"]

                    if rows_of_patients[row]["Age at Death"] != "":
                        Patient.all_patients[row].death_age = int(rows_of_patients[row]["Age at Death"])

                    if rows_of_patients[row]["Highest level of education"] != "":
                        Patient.all_patients[row].ed_lvl = rows_of_patients[row]["Highest level of education"]

                    if rows_of_patients[row]["Cognitive Status"] != "":
                        Patient.all_patients[row].cog_stat = rows_of_patients[row]["Cognitive Status"]

                    if rows_of_patients[row]["Age of onset cognitive symptoms"] != "":
                        Patient.all_patients[row].age_symp_on = int(rows_of_patients[row]["Age of onset cognitive symptoms"])

                    if rows_of_patients[row]["Age of Dementia diagnosis"] != "":
                        Patient.all_patients[row].age_diag = int(rows_of_patients[row]["Age of Dementia diagnosis"])

                    if rows_of_patients[row]["Known head injury"] != "":
                        Patient.all_patients[row].head_inj = rows_of_patients[row]["Known head injury"]

                    if rows_of_patients[row]["Thal"] != "":
                        Patient.all_patients[row].thal_score = int(rows_of_patients[row]["Thal"].split()[1])

            else:

                warnings.warn("IDs do not match.")

    @classmethod
    def instantiate_from_csv(cls, filename: str, other_file: str):
        #open csv and create list of all rows
        with open(filename, encoding="utf8") as f:
            reader = csv.DictReader(f)
            rows_of_patients = list(reader)
            #for line in csv create object
            for row in rows_of_patients:
                Patient(
                    DonorID = row["Donor ID"],
                    ABeta40 = float(row["ABeta40 pg/ug"]),
                    ABeta42 = float(row["ABeta42 pg/ug"]),
                    tTau = float(row["tTAU pg/ug"]),
                    pTau = float(row["pTAU pg/ug"])
                )
            Patient.all_patients.sort(key = Patient.get_id)
            Patient.combine_data(other_file)

    @classmethod
    def filter(cls, list, ABeta40:float =*any", ABeta42:float =*any", tTau:float= *any", pTau:float =*any", sex:str =*any", death_age:int =*any", ed_lvl:str =*any", cog_stat:str =*any", a
        all_patients = list
        remove_list = []
        attr_list = [
                    ABeta40,
                    ABeta42,
                    tTau,
                    pTau,
                    sex,
                    death_age,
                    ed_lvl,
                    cog_stat,
                    age_symp_on,
                    age_diag,
                    head_inj,
                    thal_score
                    ]
        attr_name = [
                    "ABeta40",
                    "ABeta42",
                    "tTau",
                    "pTau",
                    "sex",
                    "death_age",
                    "ed_lvl",
                    "cog_stat",
                    "age_symp_on",
                    "age_diag",
                    "head_inj",
                    "thal_score"
                    ]

        for attr in range(len(attr_list)):
            if attr_list[attr] != "any":
                for patient in all_patients:
                    if getattr(patient,attr_name[attr]) != attr_list[attr]:
                        remove_list.append(patient)
                all_patients = [patient for patient in all_patients if patient not in remove_list]
                remove_list.clear()

        return all_patients


# Main execution code
Patient.instantiate_from_csv('UpdatedLuminex.csv', 'UpdatedMetaData.csv')

print("Total patients loaded:", len(Patient.all_patients))

# Filter for patients who have dementia diagnosis and age of diagnosis data
patients_with_diagnosis = [p for p in Patient.all_patients if p.age_diag is not None and p.head_inj is not None]

print(f"Patients with diagnosis age and head injury data: {len(patients_with_diagnosis)}")

# Separate patients by head injury status
age_diag_head_inj_yes = []
age_diag_head_inj_no = []

for patient in patients_with_diagnosis:
    if patient.head_inj == "Yes":
        age_diag_head_inj_yes.append(patient.age_diag)
    elif patient.head_inj == "No":
        age_diag_head_inj_no.append(patient.age_diag)

print(f"Patients with head injury (Yes): {len(age_diag_head_inj_yes)}")
print(f"Patients with no head injury (No): {len(age_diag_head_inj_no)}")

if len(age_diag_head_inj_yes) > 0 and len(age_diag_head_inj_no) > 0:
    # Calculate statistics
    mean_age_diag_yes = statistics.mean(age_diag_head_inj_yes)
    mean_age_diag_no = statistics.mean(age_diag_head_inj_no)

    stdev_age_diag_yes = statistics.stdev(age_diag_head_inj_yes) if len(age_diag_head_inj_yes) > 1 else 0
    stdev_age_diag_no = statistics.stdev(age_diag_head_inj_no) if len(age_diag_head_inj_no) > 1 else 0

    print(f"Head injury Yes - Mean age at diagnosis: {mean_age_diag_yes:.2f}, Std dev: {stdev_age_diag_yes:.2f}")
    print(f"Head injury No - Mean age at diagnosis: {mean_age_diag_no:.2f}, Std dev: {stdev_age_diag_no:.2f}")

    # Statistical test
    if len(age_diag_head_inj_yes) > 1 and len(age_diag_head_inj_no) > 1:
        t_stat, p_val = stats.ttest_ind(age_diag_head_inj_yes, age_diag_head_inj_no)
        print(f"t-statistic: {t_stat:.4f}")
        print(f"p-value: {p_val:.4f}")

    # Create the bar chart
    head_inj_categories = ['Yes', 'No']
    mean_ages = [mean_age_diag_yes, mean_age_diag_no]
    stdev_ages = [stdev_age_diag_yes, stdev_age_diag_no]

    plt.figure(figsize=(10, 6))
    bars = plt.bar(head_inj_categories, mean_ages, yerr=stdev_ages, capsize=10,
                   color=["red", "blue"], alpha=0.7, edgecolor='black')

    plt.title("Age of Diagnosis by Known Head Injury", fontsize=14, fontweight='bold')
    plt.xlabel("Known Head Injury", fontsize=12)
    plt.ylabel("Age of Diagnosis", fontsize=12)
    plt.ylim(0, max(mean_ages) + max(stdev_ages) + 10)

    # Add grid for better readability
    plt.grid(axis='y', alpha=0.3, linestyle='--')

    plt.tight_layout()
    plt.show()

else:
    print("Insufficient data to create meaningful comparison")
    print("Age diagnosis data available for head injury groups:")
    print(f"Yes: {age_diag_head_inj_yes}")
    print(f"No: {age_diag_head_inj_no}")
```

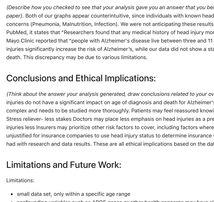
```
Total patients loaded: 84
Patients with diagnosis age and head injury data: 13
Patients with head injury (Yes): 6
Patients with no head injury (No): 7
Head injury Yes - Mean age at diagnosis: 82.00, Std dev: 11.51
Head injury No - Mean age at diagnosis: 78.57, Std dev: 11.79
t-statistic: 0.4813
p-value: 0.6308
```

Scatterplot age of death and years of education + export data as AgeEducationData.csv + regression plot

In [15]:
```
# AGE OF DEATH VS YEARS OF EDUCATION SCATTERPLOT
import pandas as pd
import matplotlib.pyplot as plt

# Load the dataset
file_path = "UpdatedMetaData.csv"  # adjust if needed
df = pd.read_csv(file_path)

# Extract relevant columns and drop rows with missing values
plot_data = df[['Age at Death', 'Years of education']].dropna()

# Create scatter plot
plt.figure(figsize=(8,6))
plt.scatter(plot_data['Years of education'], plot_data['Age at Death'], alpha=0.6)
plt.title("Scatterplot of Age at Death vs. Years of Education")
plt.xlabel("Years of Education")
plt.ylabel("Age at Death")
plt.grid(True, linestyle='--', alpha=0.5)
plt.show()


# CSV EXPORT
# Load dataset
file_path = "UpdatedMetaData.csv"  # adjust if needed
df = pd.read_csv(file_path)

# Select relevant columns and drop missing values
export_data = df[['Age at Death', 'Years of education']].dropna()

# Save to new CSV file
export_data.to_csv("AgeEducationData.csv", index=False)

print("Data exported to AgeEducationData.csv")

#REGRESSION
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score

df = pd.read_csv("AgeEducationData.csv")
x = df["Age at Death"].values.reshape(-1, 1)
y = df["Years of education"].values

model = LinearRegression()
model.fit(x, y)
slope = model.coef_[0]
intercept = model.intercept_
r2 = model.score(x, y)

plt.scatter(x, y, color='blue')
plt.plot(x, model.predict(x), color="red")
# Annotate equation
equation = f"y = {slope:.2f}x + {intercept:.2f}\nR² = {r2:.2f}"
plt.text(x.min(), y.max(), equation, color="red", fontsize=12, verticalalignment='top')
# Annotate scatterplot with labels and title
plt.xlabel("Age at Death")
plt.ylabel("Years of education")
plt.title("Age at Death vs. Years of education")
plt.show()
```




```
Data exported to AgeEducationData.csv
```



Verify and validate your analysis:

(Describe how you checked to see that your analysis gave you an answer that you believe (verify). Describe how you determined if your analysis gave you an answer that is supported by other evidence (e.g., a published paper). Both of our graphs appear counterintuitive, since individuals with known head injuries are generally considered more likely to develop Alzheimer's as well as experience earlier death due to Alzheimer's-related health concerns (Pneumonia, Malnutrition, Infection). We were not anticipating these results because we already did research on the impact head injuries have on Alzheimer's disease patients. According to an article published on PubMed, it states that "Research has found that an individual's risk of developing Alzheimer's Disease and the chance of developing non-Alzheimer's dementia". Additionally, the Mayo Clinic reported that "people with Alzheimer's disease live between three and 11 years after diagnosis" When we compare our findings to published research, the results are not consistent. Prior studies suggest that head injuries significantly increase the risk of Alzheimer's, while our data did not show a statistically significant difference in the age of diagnosis between those with and without head injuries, nor did it reveal a difference in age at death. This discrepancy may be due to various limitations.

Conclusions and Ethical Implications:

(Think about the answer your analysis generated, draw conclusions related to your overarching question, and discuss the ethical implications of your conclusions. Based on graphs, we can conclude that previous known head injuries do not have a significant impact on age of diagnosis and death for Alzheimer's disease patients. Results contradict research, proving that the relationship between head injuries and Alzheimer's disease progression is complex and needs to be studied more thoroughly. Patients may feel reassured knowing that head injuries do not appear to significantly affect the age of diagnosis or overall lifespan in individuals with Alzheimer's disease. Stress relever- less stakes Doctors may place less emphasis on head injuries as a predictor of Alzheimer's disease, since they do not appear to greatly influence Alzheimer's-related mortality. Hospitals may screen for head injuries less Insurers may prioritize other risk factors to cover, including factors where we can find concrete data evidence that suggests there is a direct correlation between the risk factor and head injuries. It would be unjustified for insurance companies to use head injury status to determine insurance when it does not significantly impact Alzheimer's diagnosis or death. ** approach all these claims with caution due to the inconsistency we had with research and data results. These are all ethical implications based on the data we analyzed.

Limitations and Future Work:

Limitations:

- small data set, only within a specific age range
- confounding variables such as APOE genes or other health concerns may have skewed the data
  - this is why we can't prove causation from correlation
- limited head injury information - just the yes or no

Future work: Our original idea was to compare the gender, head injuries, and alzheimer's disease progression. We quickly realized that these were too many variables, so we eliminated the gender portion. If we were to continue with this research, we would love to incorporate it back in. We are interested to see how sex affects whether or not someone is more likely to have a known head injury, and how it will consequently affect Alzheimer's disease progression. We want to look at a wider data set next time in order to get more accurate results that may align better with our research. We also want to look into how severe their previous head injuries actually were and impact of repeated head injuries. This requires more data collection but would allow us to make more graphs and draw more connections.

Class Notes:

Brainstorm questions:
Are specific races more prone to developing alzheimer's disease? How do level of education and occupation increase the risk of alzheimer's disease? If you have an APOE gen, are you more prone to an earlier death? Do previous head injuries lead to an increased risk of alzheimer's disease? How does this vary and compare between men and women? Do men or women die earlier from Alzheimer's (after they are diagnosed)? If you don't have easy access to healthcare, does the disease progress faster? We picked "Do previous head injuries lead to an increased risk of alzheimer's disease? How does this vary and compare between men and women?" to do our project on. We decided that our question was too complex with too many variables, so we decided to eliminate the gender portion and just focus on head injuries as it relates to alzheimers progression and death due to the disease.

Questions for TA:

None at the moment :)